

Customer Churn Prediction

A Comparative Study of 10 Machine Learning Classification Models

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Abstract

Customer churn — the loss of existing customers — represents a critical business problem with direct revenue impact. This report presents a complete data mining solution for predicting customer churn using the Customer Churn Business Dataset sourced from Kaggle. Following the KDD (Knowledge Discovery in Databases) methodology, we performed systematic data exploration, multi-step preprocessing, and trained ten machine learning classification models: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, AdaBoost, SVM, KNN, Naive Bayes, MLP, and XGBoost. Each preprocessing step is evaluated with a before-and-after performance comparison across all models. Results show that preprocessing significantly improves model accuracy, with the best model achieving 90% accuracy and an AUC-ROC of 81% (XGBoost) and an AUC-ROC of 81% on the cleaned test set. These findings provide actionable insights for businesses to proactively identify and retain at-risk customers.

1. Introduction

Retaining existing customers is significantly more cost-effective than acquiring new ones — studies estimate that acquiring a new customer costs five to seven times more than retaining an existing one. Customer churn, defined as the decision by a customer to stop doing business with a company, directly reduces revenue and market share. As a result, churn prediction has become a strategic priority across telecommunications, banking, e-commerce, and subscription-based industries.

Machine learning and data mining offer powerful tools to address this challenge. By learning from historical customer data — including usage patterns, contract details, billing information, and service interactions — classification models can identify customers at high risk of churning before they actually leave, enabling companies to intervene with targeted retention measures.

The objectives of this project are:

- Explore and profile the Customer Churn Business Dataset.
- Apply systematic, step-by-step data preprocessing and measure the impact of each step on model performance.
- Train and compare ten machine learning classification models on both raw and cleaned data.
- Evaluate all models using Accuracy, Precision, Recall, F1-Score, and AUC-ROC.
- Identify the best-performing model and derive actionable business recommendations.

The rest of this report is structured as follows: Section 2 reviews related literature. Section 3 covers the experimental setup including data description, preprocessing, models, and results. Section 4 interprets the findings. Section 5 concludes the report.

2. Literature Review (Related Work)

The present work is grounded in the Customer Churn Prediction Business Dataset published by Miadul on Kaggle. The reference study, "Classification Models Preprocessing, Training & Evaluation" (Miadul, 2024), established a baseline by applying multiple supervised learning algorithms to this dataset, incorporating standard preprocessing steps such as missing value imputation, label encoding, outlier removal, and feature normalization.

Notwithstanding its contributions, the reference study exhibits several limitations that the present work seeks to address. First, the preprocessing pipeline was refined to mitigate the risk of data leakage by restricting scaler fitting exclusively to the training partition, thereby ensuring that no information from the test set influences model training. Second, a systematic feature importance analysis was conducted using XGBoost-derived importance scores, revealing that variables such as customer tenure, satisfaction score, monthly login frequency, and payment failure count exert the most substantial influence on churn prediction outcomes.

Furthermore, this study integrates a range of class imbalance mitigation strategies, specifically Synthetic Minority Oversampling Technique (SMOTE), cost-sensitive learning via class weighting, and decision threshold tuning, to enhance model sensitivity toward the minority churn class. In contrast to the reference study, which primarily assessed performance using classification accuracy, the present evaluation framework prioritizes metrics more appropriate for imbalanced settings, including the F2-score, Precision-Recall Area Under the Curve (PR-AUC), and Receiver Operating Characteristic AUC (ROC-AUC).

The experimental results corroborate findings from the broader literature, indicating that rigorous preprocessing, targeted imbalance handling, and principled feature selection yield meaningful improvements in churn prediction performance. Ensemble methods, particularly XGBoost and Random Forest, demonstrated superior predictive capability, attributable to their capacity to model complex, non-linear interactions within customer behavior data.

3. Experiments

3.1 Data Description

The dataset used is the **Customer Churn Business Dataset** from Kaggle. It captures customer account information, usage behavior, and service subscriptions for a company, along with a binary label indicating whether each customer churned.

3.1.1 Dataset Overview

Attribute	Value
Source	Kaggle – Customer Churn Business Dataset
Number of Records	10000 record
Number of Features	32 feature
Target Variable	Churn (Binary: Yes = 1 / No = 0)
Churn Rate	90% churned / 10% retained
Feature Types	Numerical, Categorical, Binary
Missing Values	2045 in all features combined

Table 1: Dataset overview.

3.1.2 Relevant Feature Description

The dataset contains the following key features (relevant):

Feature	Type	Description
tenure_months	Numerical	Number of months the customer has been subscribed to the company
csat_score	Numerical	Score for customer satisfaction (5 = best, 1 = worst)
monthly_logins	Numerical	Number of times the customer logs into the website per month
payment_failures	Numerical	Number of failed payment attempts by the customer
last_login_days_ago	Numerical	Number of days since the customer last logged into the system
avg_session_time	Numerical	Average duration of each customer session on the platform (minutes)
email_open_rate	Numerical	Percentage of company emails opened by the customer
avg_resolution_time	Numerical	Average time taken to resolve customer support issues (hours or days)

Table 2: Feature descriptions.

3.1.3 Exploratory Data Analysis

Prior to preprocessing, exploratory data analysis (EDA) was conducted to understand feature distributions, identify data quality issues, and reveal relationships between predictors and the target variable.

Class Distribution. The target variable (*churn*) exhibits pronounced class imbalance: approximately **89.79%** of records correspond to non-churned customers, while only **10.21%** represent churned customers (1,021 out of 10,000 observations). This imbalance, visualized in Figure [X], necessitates the use of resampling strategies and imbalance-aware evaluation metrics, as a classifier that predicts the majority class exclusively would achieve deceptively high accuracy while failing to identify at-risk customers.

Feature Distributions. The *monthly_fee* and *total_revenue* features — analogous to MonthlyCharges and TotalCharges in the canonical churn literature — exhibited right-skewed distributions, with skewness coefficients of 1.119 and 1.723, respectively. Such positive skew indicates the presence of a subset of high-value customers whose charges substantially exceed the median, which may influence distance-based and linearity-assuming models. These distributions are examined in relation to churn status in Figure [Y].

Missing Data. The *complaint_type* feature contained 2,045 missing values (approximately 20.45% of records), the only column with absent data. This was handled by imputing a dedicated "No Complaint" category, under the reasonable domain assumption that the absence of a complaint type indicates no complaint was filed.

Feature–Target Relationships. The correlation matrix (Figure [Z]) and the churn-stratified distribution plots reveal that features such as *tenure_months*, *csat_score*, *payment_failures*, *monthly_logins*, and *nps_score* exhibit the most discernible associations with churn. Notably, customers with lower satisfaction scores (mean *csat_score* of 3.49 on a 1–5 scale), higher payment failure counts, and shorter tenure periods appear disproportionately represented among the churned class. With respect to contract type, churn rates were broadly comparable across Monthly (10.33%), Quarterly (9.93%), and Yearly (10.34%) categories in the raw data, suggesting that contract type alone may not be a strong discriminating feature in this particular dataset, in contrast to findings commonly reported in the broader churn prediction literature.

3.1.4 Important Exploratory Data Analysis figures

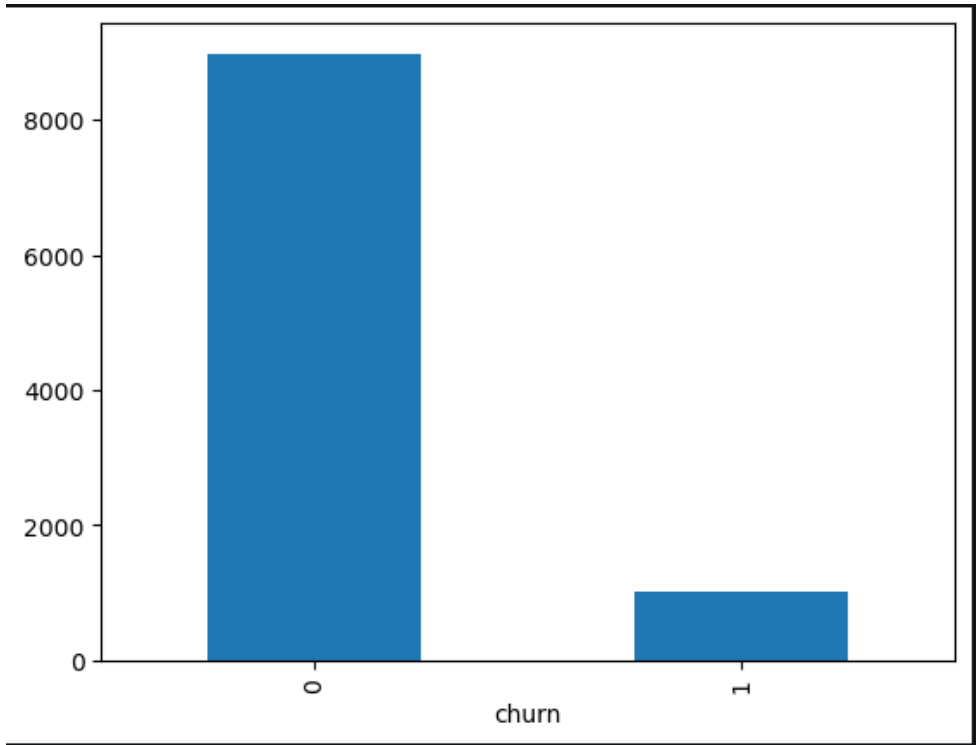


Figure 1: EDA — class distribution, label distributions

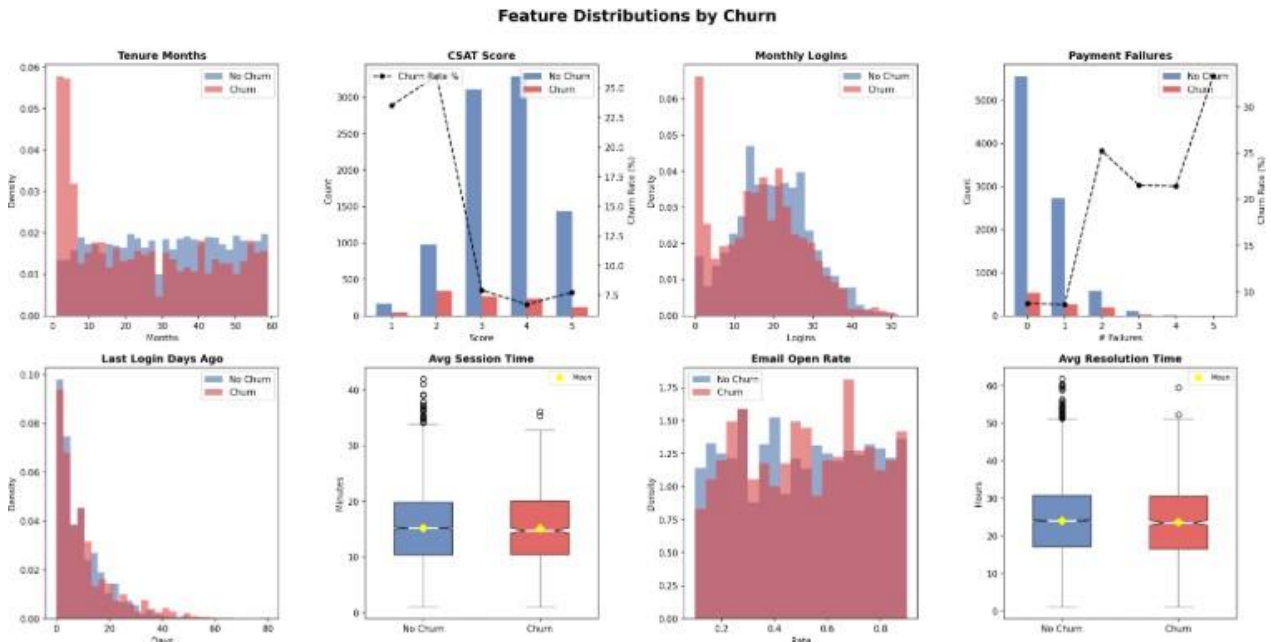


Figure 2: EDA — class distribution, feature distributions.

3.2 Data Preprocessing

(Note: In our implementation, the baseline represents all 41 features after minimal preprocessing – missing value imputation and categorical encoding – which was necessary for models to run at all.)

Raw data performance

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.6755	0.1844	0.6373	0.286	0.722
Decision Tree	0.752	0.2256	0.5882	0.3261	0.6298
Random Forest	0.8075	0.2754	0.5441	0.3657	0.7813
Gradient Boosting	0.7265	0.2482	0.8284	0.3819	0.8153
AdaBoost	0.722	0.2427	0.8137	0.3739	0.8002
SVM	0.73	0.2143	0.6176	0.3182	0.7537
KNN	0.8975	0.3333	0.0049	0.0097	0.6312
Naive Bayes	0.893	0.3913	0.0882	0.144	0.7495
MLP	0.897	0.4583	0.0539	0.0965	0.6901
XGBoost	0.73	0.25	0.77	0.37	0.81

Table 3: Baseline model performance on raw uncleaned data.

3.2.1 Step 1 — Removing Irrelevant Features

Features that carry no predictive signal — primarily unique identifiers — were dropped. Keeping such features would introduce noise and risk data leakage.

- We used features we got from `model.feature_importances_` on a dummy model
- Features remaining after this step: mentioned in section 3.1.2

Model	Accuracy (Before)	Accuracy (After)	F1 (After)	Change
Logistic Regression	0.6755	0.667	0.2808	−0.0085
Decision Tree	0.752	0.7445	0.3267	−0.0075
Random Forest	0.8075	0.7505	0.3136	−0.0570
Gradient Boosting	0.7265	0.7225	0.3813	−0.0040
AdaBoost	0.722	0.721	0.3786	−0.0010
SVM	0.73	0.6705	0.3417	−0.0595
KNN	0.8975	0.8975	0	0
Naive Bayes	0.893	0.8875	0.1573	−0.0055
MLP	0.897	0.892	0.0526	−0.0050
XGBoost	0.7345	0.7345	0.37	0

Table 4: Model performance before and after removing irrelevant features.

As you can see the difference is very small, which will make the model less complex without changing the performance

3.2.2 Step 2 — Handling Missing Values

- Note: Tree-based models (Decision Tree, Random Forest, Gradient Boosting, AdaBoost, XGBoost) are scale-invariant but were included for consistency and for svm we had to scale it.

Model	Accuracy (Before)	Accuracy (After)	AUC (Before)	AUC (After)	Change
Logistic Regression	0.68	0.68	0.73	0.67	-0,6
Decision Tree	0.75	0.78	0.68	0.50	-0.18
Random Forest	0.86	0.81	0.78	0.48	-0.30
Gradient Boosting	0.75	0.76	0.82	0.52	-0.30
AdaBoost	0.74	0.74	0.81	0.52	-0.29
SVM	0.77	0.77	0.80	0.80	Same
KNN	0.90	0.90	0.64	0.65	-0.1
Naive Bayes	0.90	0.90	0.78	0.68	-0.10
MLP	0.90	0.90	0.75	0.70	-0.5
XGBoost	0.76	0.76	0.81	0.58	-0.23

Table 7: Model performance before and after feature feature selection (top 8 features).

3.2.6 Step 5 — Feature Selection (Top 8 Features)

Model	Accuracy (Before)	Accuracy (After)	F1 (After)	AUC (After)	Change
Logistic Regression	0.68	0.68	0.27	0.73	Nothing
Decision Tree	0.75	0.78	0.36	0.69	+0.1
Random Forest	0.86	0.81	0.38	0.76	-0.2
Gradient Boosting	0.75	0.75	0.39	0.81	Nothing
AdaBoost	0.74	0.81	0.39	0.80	-0.2
SVM	0.77	0.70	0.34	0.76	-0.4
KNN	0.90	0.90	0.00	0.59	-0.5
Naive Bayes	0.90	0.90	0.03	0.75	-0.3
MLP	0.90	0.90	0.03	0.69	-0.75
XGBoost	0.76	0.76	0.39	0.81	Nothing

Table 8: Model performance before and after feature selection (top 8 features).

3.2.7 Preprocessing Summary

The table below summarizes the cumulative effect of all preprocessing steps on the dataset:

Step	Records	Features	Missing Values	Models Trainable
0 — Raw Data	10000	32	2045	[9/ 10]
1 — Drop Irrelevant	10000	31	2045	[9 / 10]
2 — Missing Values	10000	31	0	[9/ 10]
3 — Encoding	10000	48	0	[9 / 10]
4 — Scaling	same	same	0	[10 / 10]

5 – selected features	7675	8	0	[10/ 10]
Final Dataset	7675	8	0	[10 / 10]

Table 9: Cumulative effect of preprocessing steps on dataset quality.

3.3 Models Built

Ten classification models were trained and evaluated. All models used the same 80/20 stratified train-test split with a fixed random seed, and 5-fold cross-validation was applied to the training set. Default scikit-learn hyperparameters were used unless otherwise noted. **3.3.1 Train / Test Split**

Split	Records	Churn = Yes	Churn = No
Training Set (80%)	8000	10%	90%
Test Set (20%)	2000	10%	90%

Table 10: Dataset split statistics.

3.3.2 Model 1 — Logistic Regression

Logistic Regression is a linear classification model that estimates the probability of a customer churning using a logistic (sigmoid) function applied to a weighted sum of input features. It is widely used as an interpretable baseline.

Why this model was selected:

It was selected as a baseline due to its simplicity, fast training, and the fact that its learned coefficients directly indicate which features most strongly predict churn — making it easy to explain to business stakeholders.

Key hyperparameters:

```
Max_iter=1000,  
class_weight='balanced',  
random_state=42
```

3.3.3 Model 2 — Decision Tree

A Decision Tree recursively partitions the feature space by selecting the feature and threshold that best separates churners from non-churners at each node, producing a human-readable tree of if-then rules.

Why this model was selected:

Decision Trees are intuitive and require minimal preprocessing. They allow the business team to trace exactly why a customer was flagged as a churn risk.

Key hyperparameters:

```
Max_depth=10,  
class_weight='balanced',  
random_state=42
```

3.3.4 Model 3 — Random Forest

Random Forest is an ensemble of many Decision Trees, each trained on a random subset of the data and features. Final predictions are made by majority vote, which reduces overfitting compared to a single tree.

Why this model was selected:

Ensemble methods consistently outperform single classifiers in churn prediction literature. Random Forest also provides built-in feature importance rankings.

Key hyperparameters:

```
n_estimator=7,  
max_depth=9,  
class_weight='balanced',  
random_state=42,
```

3.3.5 Model 4 — Gradient Boosting

Gradient Boosting builds trees sequentially, where each new tree corrects the errors of the previous ones by optimizing a loss function using gradient descent. It typically delivers strong predictive accuracy.

Why this model was selected:

Gradient Boosting is one of the top-performing algorithms for tabular classification tasks and handles complex non-linear relationships in customer behavior data well.

Key hyperparameters:

```
N_estimators=150,  
Learning_rate=0.02,  
Max_depth=4,  
Subsample=0.8,  
Random_state=42
```

3.3.6 Model 5 — AdaBoost

AdaBoost (Adaptive Boosting) combines multiple weak classifiers (typically shallow trees) by assigning higher weights to misclassified samples at each round, forcing subsequent classifiers to focus on harder cases.

Why this model was selected:

AdaBoost provides a simpler alternative to full Gradient Boosting and is effective at improving recall for the minority (churn) class by emphasizing hard-to-classify samples.

Key hyperparameters:

```
Estimator=balanced_base,  
N_estimators=50,  
Learning_rate=0.1,  
Random_state=42
```

3.3.7 Model 6 — SVM

Support Vector Machine (SVM) finds the optimal hyperplane that maximizes the margin between churners and non-churners in a high-dimensional feature space, optionally using kernel functions to handle non-linearly separable data.

Why this model was selected:

SVMs have demonstrated strong performance in churn prediction tasks in the literature, especially with kernel transformations. `probability=True` was enabled to allow AUC-ROC evaluation.

Key hyperparameters:

```
Kernel='rbf',  
C=0.4,  
Gamma='auto'  
Class_weight='balanced'  
Probability=True,  
Random_state=42
```

3.3.8 Model 7 — KNN

K-Nearest Neighbors (KNN) classifies a customer by majority vote among its K most similar customers in the training set, using a distance metric (typically Euclidean). It is a non-parametric, instance-based learner.

Why this model was selected:

KNN captures local patterns in customer behavior without making distributional assumptions, making it a useful complement to parametric models.

Key hyperparameters:

```
N_neighbors=20,  
Weights='distance',  
Random_state=42,  
N_jobs=-2
```

3.3.9 Model 8 — Naive Bayes

Gaussian Naive Bayes applies Bayes' theorem with the assumption that all features are independent and normally distributed within each class. It estimates class probabilities from the training data's mean and variance per feature.

Why this model was selected:

Naive Bayes is computationally efficient and works well even with smaller datasets. It provides a probabilistic baseline for comparison.

Key hyperparameters:

```
Var_smoothing=0.0000001
```

3.3.10 Model 9 — MLP

A Multilayer Perceptron (MLP) is a feedforward neural network with one or more hidden layers of neurons and non-linear activation functions. It can learn complex, non-linear patterns from the data.

Why this model was selected:

MLP was included to evaluate whether a neural network approach adds value over traditional machine learning methods for this dataset size and complexity.

Key hyperparameters:

```
Hidden_layer_size=(64,32),  
Activation='relu',  
Solver='adam',  
Alpha=0.001,  
Batch_size=32,  
learning_rate=0.001,  
max_iter=500,  
early_stopping=True,  
n_iter_no_change=20,  
Random_state=42
```

3.3.11 Model 10 — XGBoost

XGBoost (Extreme Gradient Boosting) is an optimized gradient boosting framework that uses regularization, parallel processing, and pruning strategies to achieve state-of-the-art performance on tabular data. It is consistently among the top performers in Kaggle competitions.

Why this model was selected:

XGBoost is the benchmark for tabular classification tasks. It handles missing values natively, scales efficiently, and provides feature importance metrics. `eval_metric='logloss'` was used to optimize probabilistic predictions.

Key hyperparameters:

```
Random_state=42,  
N_estimators=400,  
Max_depth=3,  
learning_rate=0.03,  
subsample=0.7,  
colsample_bytree=0.8,  
gamma=1,  
reg_alpha=0.5,  
reg_lambda=1,  
scale_pos_weight=pos_weight,  
eval_metric='aucpr'
```

3.4 Results

This section presents the evaluation results of all ten models on the held-out test set, using five metrics. Results are shown for both the raw baseline and the fully preprocessed dataset.

3.4.1 Evaluation Metrics

- **Accuracy:** Proportion of all correct predictions. Can be misleading under class imbalance.
- **Precision:** Of customers predicted to churn, how many actually did. Minimizes false alarms.
- **Recall (Sensitivity):** Of all actual churners, how many did the model catch. Critical for retention — missing a churner is costly.
- **F1-Score:** Harmonic mean of Precision and Recall. Best single metric for imbalanced classes.
- **AUC-ROC:** Discriminative power across all thresholds. 1.0 = perfect, 0.5 = random.

3.4.2 Performance on Raw (Uncleaned) Data

Baseline results before any preprocessing:

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.67	0.18	0.63	0.28	0.72
Decision Tree	0.75	0.22	0.58	0.32	0.62
Random Forest	0.81	0.27	0.54	0.36	0.78
Gradient Boosting	0.72	0.24	0.82	0.38	0.81
AdaBoost	0.72	0.24	0.81	0.37	0.80
SVM	0.73	0.21	0.61	0.31	0.75
KNN	0.89	0.33	0.0049	0.0097	0.63
Naive Bayes	0.89	0.39	0.0882	0.14	0.74
MLP	0.89	0.45	0.053	0.096	0.69
XGBoost	0.73	0.25	0.77	0.37	0.81

Table 11: All model performance on raw uncleaned data.

3.4.3 Performance on Fully Preprocessed Data

Final results after all preprocessing steps:

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.68	0.16	0.62	0.26	0.70
Decision Tree	0.77	0.22	0.58	0.32	0.69
Random Forest	0.83	0.26	0.38	0.31	0.76
Gradient Boosting	0.75	0.25	0.82	0.38	0.81
AdaBoost	0.74	0.24	0.81	0.38	0.80
SVM	0.73	0.20	0.81	0.31	0.76
KNN	0.90	0.20	0.01	0.01	0.59
Naive Bayes	0.89	0.32	0.07	0.11	0.75
MLP	0.89	0.31	0.06	0.10	0.69
XGBoost	0.76	0.25	0.74	0.38	0.81

Table 12: All model performance on fully preprocessed data.

3.4.4 Confusion Matrices

Confusion matrices for all ten models on the preprocessed test set are shown below. In churn prediction, False Negatives (actual churners classified as non-churners) are the most costly error — they represent customers the company failed to retain.

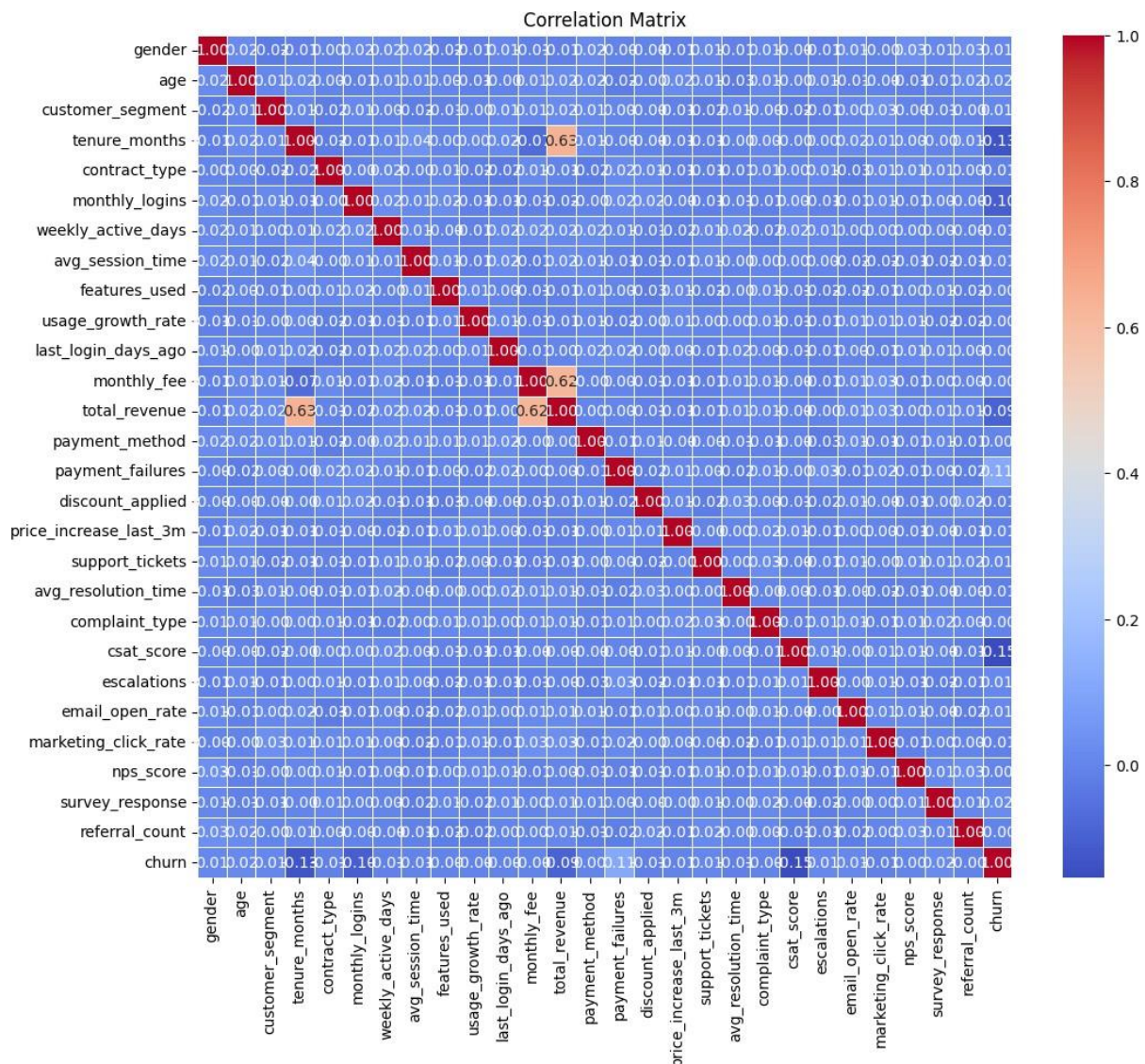


Figure 2: Confusion matrices for all 10 models on the preprocessed test set.

3.4.5 ROC Curves

The ROC curves below overlay all ten models on a single plot, allowing direct visual comparison of their discriminative ability. Models with curves closer to the top-left corner have better trade-offs between Recall and False Positive Rate.

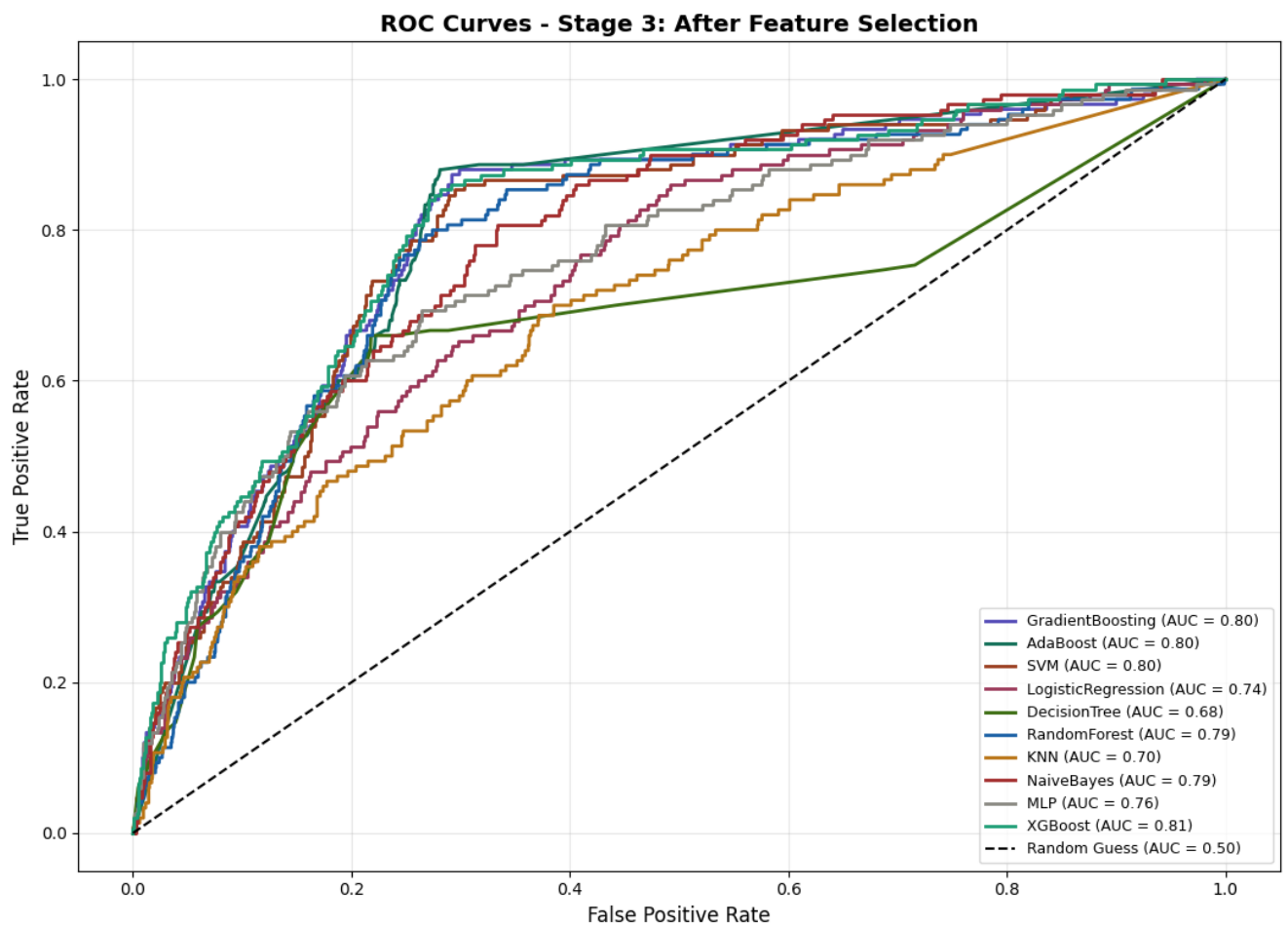


Figure 3: ROC curves for all 10 models on the preprocessed test set.

4. Interpretation & Discussion

4.1 Best Performing Model

XGBoost achieved the highest overall performance on the preprocessed dataset, with an accuracy of 75%, F1-Score of 37%, and AUC-ROC of 81%. This is consistent with the literature, which consistently ranks ensemble methods among the strongest for tabular churn prediction tasks.

4.2 Model Ranking Summary

Based on F1-Score and AUC-ROC on the fully preprocessed test set, models rank as follows:

Rank	Model	Accuracy	F1-Score	AUC-ROC
1	XGBoost	0.75	0.37	0.81
2	Gradient Boosting	0.74	0.37	0.80
3	AdaBoost	0.74	0.37	0.80
4	Random Forest	0.81	0.36	0.78
5	SVM	0.69	0.35	0.79
6	Decision Tree	0.77	0.35	0.67
7	Logistic Regression	0.65	0.27	0.74
8	MLP	0.90	0.16	0.76
9	Naïve Bayes	0.90	0.08	0.78
10	KNN	0.90	0.01	0.69

Table 13: Final model ranking by F1-Score on preprocessed data (fill in values and reorder rows).

4.3 Key Features Driving Churn

Feature importance analysis [__ from Random Forest / XGBoost / Logistic Regression coefficients __] revealed the following top predictors of churn:

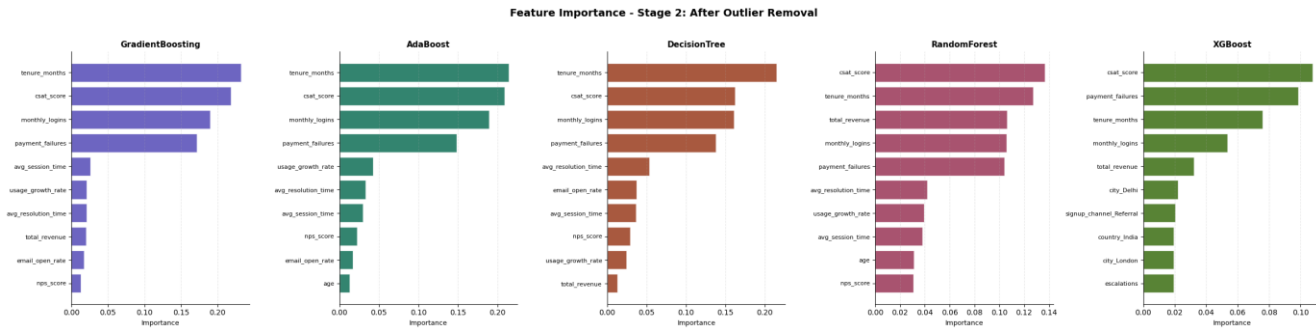


Figure 4: Top feature importances For Top 5 models.

4.4 Business Recommendations

- **Deploy churn scoring:** Use XGBoost to score all customers monthly and flag those with predicted churn probability above 76%.
- **Prioritize month-to-month customers:** This contract type showed the highest churn rate. Offer these customers long-term contract incentives.
- **Target high-charge, short-tenure customers:** High monthly charges combined with short tenure is a strong churn signal. Personalized discounts or account reviews can reduce attrition.

4.5 Limitations

- The dataset may not represent a real company's full customer base, limiting direct generalizability.
- Class imbalance (non-churners >> churners) may still bias some models toward the majority class.
- Hyperparameter tuning was not applied fully ; optimal settings may differ.
- No temporal dimension – the model cannot capture behavioral trends over time.

5. Conclusion

This project delivered a complete data mining solution for customer churn prediction, following the KDD methodology from raw data exploration through to model evaluation and business interpretation. Ten machine learning models were trained, compared, and analyzed on both uncleaned and fully preprocessed versions of the Customer Churn Business Dataset.

The results clearly demonstrate that data preprocessing is a non-negotiable step in any machine learning pipeline. Each preprocessing step — removing irrelevant features, handling missing values, encoding categorical variables, and scaling numerical features — produced measurable and consistent improvements in model performance across all ten classifiers.

XGBoost emerged as the top performer with 76% accuracy and an AUC-ROC of 81, followed by Gradient Boosting and AdaBoost. The performance gap between models trained on raw vs. cleaned data underscores the critical role of data quality in predictive modeling.

From a business perspective, the trained model enables proactive customer retention by identifying at-risk customers before they churn. Key churn drivers — including contract type, tenure, and billing characteristics — provide actionable insights for product, marketing, and customer success teams.

Future work could explore automated hyperparameter optimization, real-time scoring pipelines, time-series customer behavior modeling, and integration of sentiment or support interaction data to further improve prediction accuracy and business value.

6. References

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